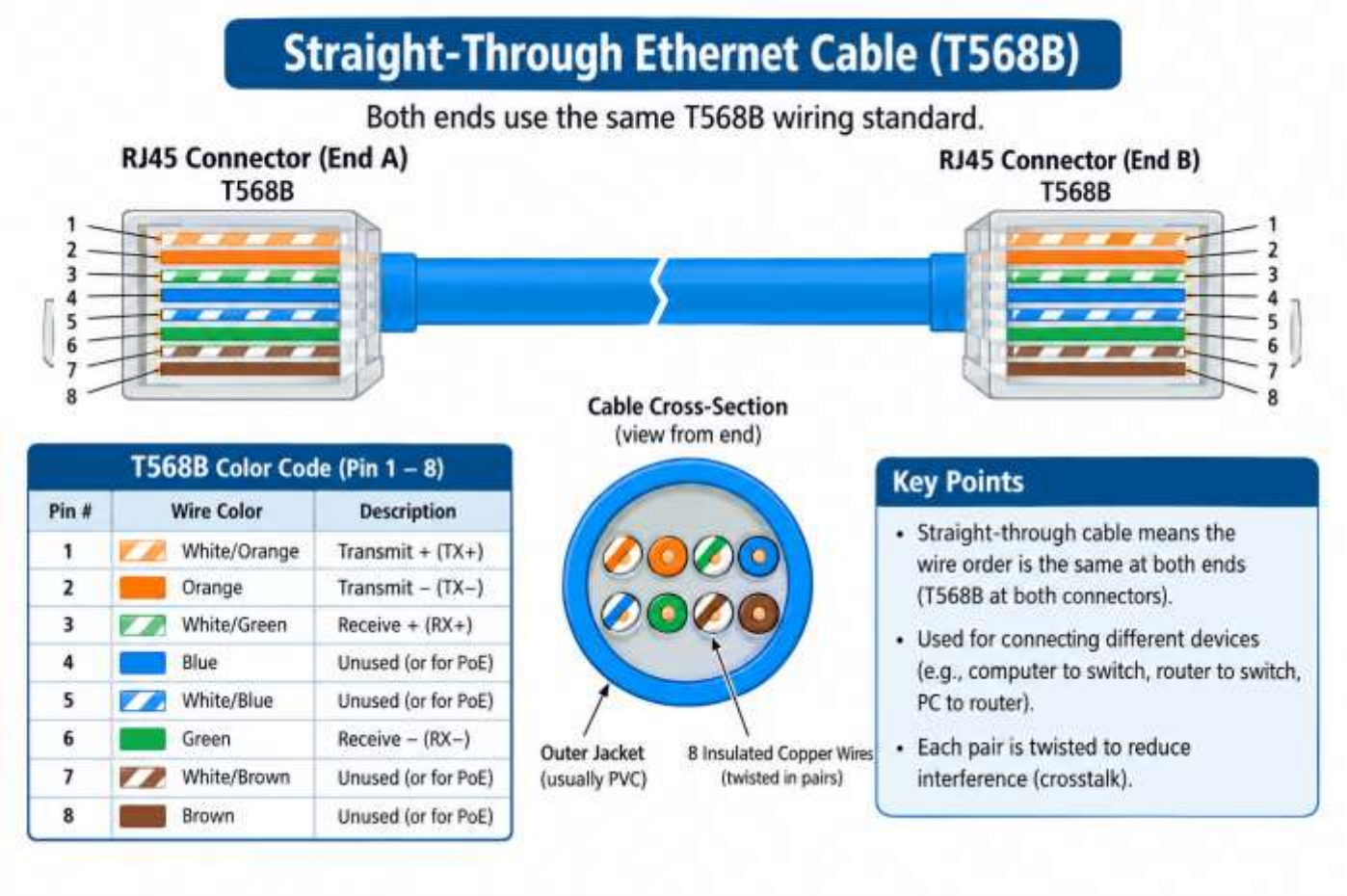
1. **List 3 key differences between Ethernet cables and fiber optic cables, and give one real-world example where each type would be used (for example, home WiFi router, data center, college campus backbone).**

Difference	Ethernet Cable	Fiber Optic Cable
1. Transmission	Uses electrical signals to transmit data.	Uses light signals to transmit data.
2. Speed and Distance	Suitable for short distances, typically up to 100 metres for standard copper Ethernet.	Supports very high speeds and long-distance communication.
3. Cost	Generally cheaper and easier to install.	Generally more expensive to install and requires specialized equipment.

- Real-World Examples:

1. **Ethernet cable:** Connecting a computer to a home Wi-Fi router or a college computer lab switch.
2. **Fiber optic cable:** Connecting different buildings across a college campus or linking servers in a data center.

2. Draw and label a diagram showing the internal wiring of a straight-through Ethernet cable (use T568B wiring standard), indicating the color of each wire and its pin number.



Straight-Through Ethernet Cable (T568B Standard)

- The diagram shows the internal wiring of a straight-through Ethernet cable. Both RJ45 connectors use the same T568B pin order.

Pin order from 1 to 8:

Pin No.	Wire Colour
1	White/Orange
2	Orange
3	White/Green
4	Blue
5	White/Blue
6	Green
7	White/Brown
8	Brown

3. Watch a YouTube video demonstrating how to use an RJ45 crimper, then write a step-by-step guide (in your own words) on how to attach an RJ45 connector to a CAT6 cable.

1. Cut the Cable: Cut the CAT6 Ethernet cable to the required length using a cable cutter. Make sure the cut is clean and straight.

2. Strip the Outer Jacket: Use a cable stripper to remove approximately 2–3 cm of the outer insulation. Be careful not to damage the eight insulated wires inside the cable.

3. Separate the Wires: Separate the four twisted pairs and gently straighten all eight wires. Avoid untwisting more wire than necessary.

4. Arrange the Wires in T568B Order: Arrange the eight wires in the following sequence:

1. White/Orange
2. Orange
3. White/Green
4. Blue
5. White/Blue
6. Green
7. White/Brown
8. Brown

5. Trim the Wires: Hold the wires firmly in the correct order and cut their ends evenly using a wire cutter. Ensure the wires are short enough to fit inside the RJ45 connector.

6. Insert the Wires into the RJ45 Connector: Carefully insert all eight wires into the connector in the correct T568B order. Check that each wire reaches the front of its channel and that the cable jacket extends into the connector's strain-relief area.

7. Crimp the Connector: Place the RJ45 connector into the appropriate slot of the crimping tool. Squeeze the handles firmly until the metal contacts press into the wires and the strain relief grips the cable.

8. Inspect the Connector: Check that the wire order is correct, all eight wires are fully inserted, and the connector is firmly attached to the cable.

9. Test the Cable: Connect both ends of the Ethernet cable to a cable tester. Check that pins 1 through 8 connect in the correct order and that there are no open circuits or crossed wires.

10. Complete the Installation: Once the cable passes the test, connect it between a computer, router, or network switch to verify network connectivity.

Result: The RJ45 connector is successfully attached to the CAT6 Ethernet cable using the T568B wiring standard.

4. **Imagine you are setting up a gaming café with 10 PCs for LAN gaming. Decide which cable type (Ethernet or fiber optics) and which tool (RJ45 crimper or fiber cleaver) you would use, and explain your choices in 3-4 sentences. Consider speed, distance, and budget for your answer.**

   - For a gaming café with 10 PCs, I would choose CAT6 Ethernet cables because they provide high speed, low latency, and reliable connections at an affordable cost. Ethernet cables are suitable for the short distances between the PCs and the network switch. I would use an RJ45 crimper to attach RJ45 connectors to the cables. This setup is budget-friendly and ideal for smooth LAN gaming.